STAT 6410
Homework 6

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1. **Chapter 2, Problem 28**

   Let μ be the Lebesgue measure on $([-1,1], \mathcal{B}([-1,1]))$. For $n \geq 1$, define $f_n(x) = nI_{(0,n^{-1})}(x) - nI_{(-n^{-1},0)}(x)$ and $f(x) \equiv 0$ for $x \in [-1,1]$. Show that $f_n \to f$ a.e. μ and $\int f_n d\mu \to \int f d\mu$ but $\{f_n\}_{n\geq 1}$ is not UI.

Ans:

Proof. For each $x \in [-1,1]$, there exists some $N \in \mathbb{N}$ such that $x \notin (-n^{-1}, n^{-1})$. Thus, for all sufficiently large $n \geq N$, $f_n(x) = 0$. Thus, $\lim_{n\to\infty} f_n(x) = f(x)$ for all $x \in [-1,1]$. Since this is true for all $x \in [-1,1]$, and $\mu([-1,1]^c) = \mu(\emptyset) = 0$, $f_n \to f$ a.e.(μ).

$$\lim_{n\to\infty} \int f_n d\mu = \lim_{n\to\infty} (n\mu(0, n^{-1}) - n\mu(-n^{-1}, 0)) = 0 - 0 = 0$$

$$\int f d\mu = 0$$

$$\therefore \int f_n d\mu \to \int f d\mu$$

Let $t > 0$ be given. $\sup_{n\geq 1} \int_{|f_n|>t} |f_n| d\mu = 2$

Thus, $\{f_n\}_{n\geq 1}$ is not uniformly integrable. □

2. **Chapter 2, Problem 30**

   For $n \geq 1$, let $f_n(x) = n^{-1/2} I_{(0,n)}(x)$, $x \in \mathbb{R}$ and let $f(x) \equiv 0$. Let m denote the Lebesgue measure on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$. Show that $f_n \to f$a.e.(m) and $\{f_n\}_{n\geq 1}$ is UI, but $\int f_n dm \not\to \int f dm$

Ans:

Proof. **WTS:** $f_n \to fa.e.(m)$

$$B = \{x \in \mathbb{R} : \lim_{n\to\infty} f_n(x) \neq f(x)\}$$
$$= \{x \in \mathbb{R} : \lim_{n\to\infty} n^{-1/2} I_{(0,n)}(x) \neq 0\}$$
$$\text{Since } \lim_{n\to\infty} n^{-1/2} = 0,\ B = \emptyset$$
$$m(\emptyset) = 0 \implies f_n \to fa.e.(m)$$

WTS: $\int f_n dm \not\to \int f dm$

$\int f dm = \int 0 dm = 0$

$\lim_{n\to\infty} f_n(x) dm(x) = \lim_{n\to\infty} (m((0,n)^c) \times 0 + m((0,n)) \times n^{-1/2}) = 0 + \lim_{n\to\infty} (\frac{n}{\sqrt{n}}) = \lim_{n\to\infty} (\sqrt{n}) \neq 0.$

Thus, $\int f_n dm \not\to \int f dm.$ □

3. **Chapter 2, Problem 36**

   (a) Let $\{f_n\}_{n\geq 1} \subset L^1(\Omega, \mathcal{F}, \mu)$ such that $f_n \to 0$ in $L^1(\mu)$. Show that $\{f_n\}_{n\geq 1}$ is UI.

   (b) Let $\{f_n\}_{n\geq 1} \subset L^p(\Omega, \mathcal{F}, \mu)$, $0 < p < \infty$, such that $\mu(\Omega) < \infty$, $\{|f_n|^p\}_{n\geq 1}$ is UI, and $f_n \to^m f$. Show that $f \in L^p(\mu)$ and $f_n \to f$ in $L^p(\mu)$.

Note: I don't know how to fix the orientation. (a) Is on the next page. (b) I'm quite confused about this one, to be honest.



4. **Chapter 2, Problem 42**

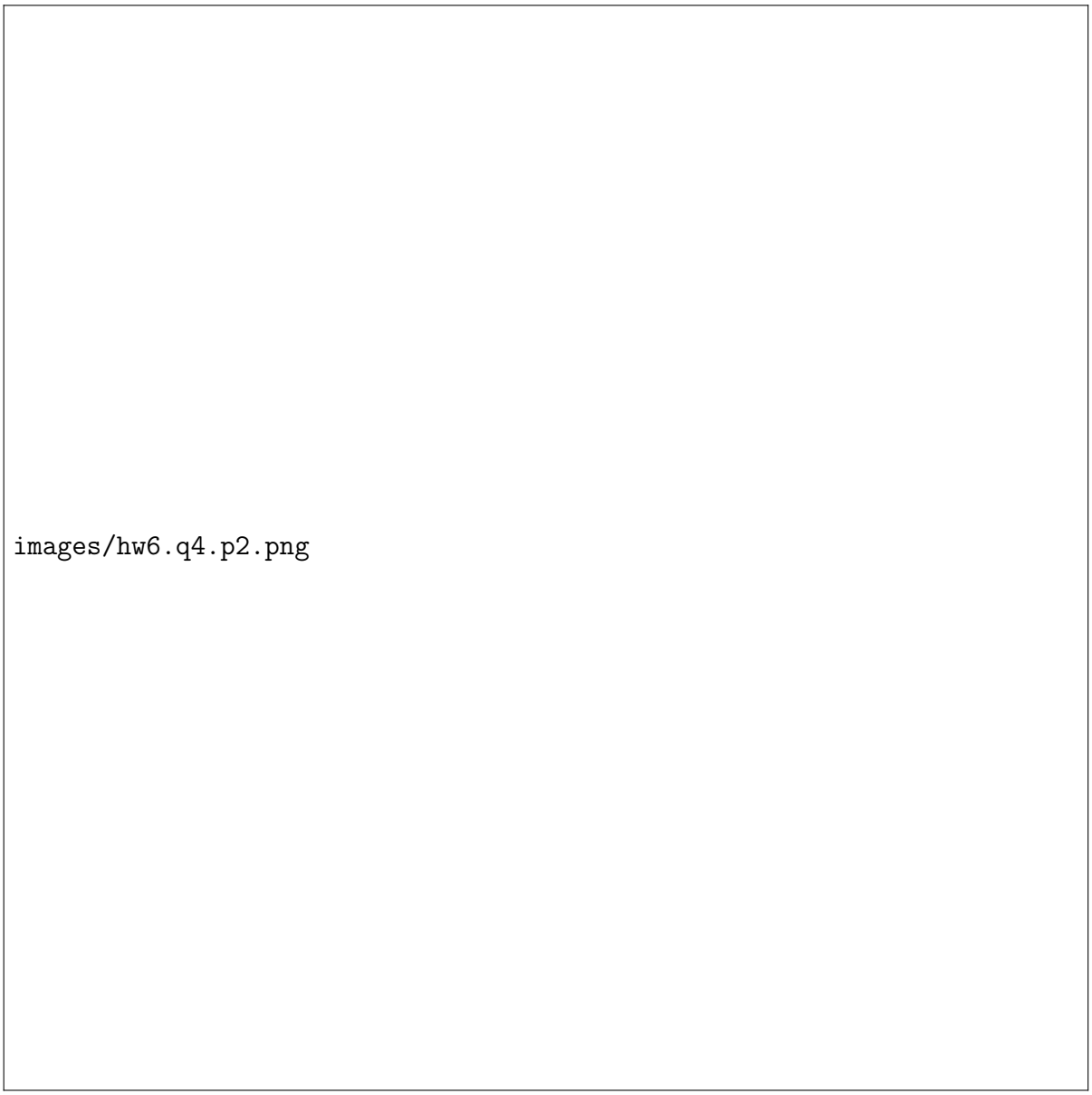
   (*Change of variables formula*) Let $(\Omega_i, \mathcal{F}_i)$, $i = 1, 2$ be two measurable spaces. Let $f : \Omega_1 \to \Omega_2$ be $\langle \mathcal{F}_1, \mathcal{F}_2 \rangle$-measurable, $h : \Omega_2 \to \mathbb{R}$ be $\langle \mathcal{F}_2, \mathcal{B}(\mathbb{R}) \rangle$-measurable, and μ_1 be a measure on $(\Omega_1, \mathcal{F}_1)$. Show that $g \equiv h \circ f$, i.e., $g(\omega) \equiv h(f(\omega))$ for $\omega \in \Omega_1$ is in $L^1(\mu)$ iff $h(\cdot) \in L^1(\Omega_2, \mathcal{F}_2, \mu_2)$ where $\mu_2 = \mu_1 f^{-1}$ iff $I(\cdot) \in L^1(\mathbb{R}, \mathcal{B}(\mathbb{R}), \mu_3 \equiv \mu_2 h^{-1})$ where $I(\cdot)$ is the identity function in $\mathbb{R}$, i.e., $I(x) \equiv x$ for all $x \in \mathbb{R}$ and also that:

$$\int_{\Omega_1} g d\mu_1 = \int_{\Omega_2} h d\mu_2 = \int_{\mathbb{R}} x d\mu_3$$

Hint: to show the first equality (and the first part of the problem), start with a simple non-negative function h and follow the usual steps to conclude about general h. The second part of the problem is related to the first part.

the work is at the bottom of this document.

images/hw6.q4.p1.png



5. **Additional Problem A:**

   Let $(\Omega, \mathcal{F}, P) = ([0,1], \mathcal{B}([0,1]), m)$, where m denotes the Lebesgue measure. Define the random variables X, $\{X_n\}$: for each $\omega \in \Omega$, $X(\omega) = 1$

$$X_1(\omega) = 0,\ X_n(\omega) = 1 + (n-1)\mathbf{I}_{\{[0,\frac{1}{n}]\}}(\omega),\ n \geq 2$$

   (a) Does $X_n \to X$ *everywhere* as $n \to \infty$?

   (b) Does $X_n \to X$ a.s.(P) as $n \to \infty$?

   (c) Does $X_n \to X$ in probability as $n \to \infty$?

(d) Does $X_n \to X$ in L^p as $n \to \infty$? Argue for different values of $p \in (0, \infty)$.

(e) Is $\{X_n\}$ a uniformly integrable sequence of random variables?

Ans:

(a) No! Below is a counterexample

Proof.

$$\text{If } \omega = 0,\ \lim_{n\to\infty} X_n(\omega) = \lim_{n\to\infty} X_n(0) = \lim_{n\to\infty} (1 + (n-1)\mathbf{I}_{[0,\frac{1}{n}]}(0))$$
$$= \lim_{n\to\infty} 1 + n - 1 = \lim_{n\to\infty} n = \infty \neq 1 = X(0)$$

□

(b) *Proof.*

$$B = \{\omega \in [0,1] : \lim_{n\to\infty} X_n \neq X\}$$
$$= \{\omega \in [0,1] : \lim_{n\to\infty} 1 + (n-1)\mathbf{I}_{[0,\frac{1}{n}]}(\omega) \neq 1\}$$
$$= \{\omega \in [0,1] : \lim_{n\to\infty} (n-1)\mathbf{I}_{[0,\frac{1}{n}]}(\omega) \neq 0\}$$
$$\text{For each } \omega \in (0,1]\ \exists N_\omega \in \mathbb{N}\ s.t.\ \omega > \frac{1}{N_\omega}$$
Clearly, as $n \to \infty$ n will surpass N_ω
When $n > N_\omega$, $\omega \notin [0, \frac{1}{n}] \implies \mathbf{I}_{[0,\frac{1}{n}]}(\omega) = 0$
$$\therefore \lim_{n\to\infty} (n-1)\mathbf{I}_{[0,1]}(\omega) = 0\ \forall\, \omega \in (0,1]$$
$$\text{If } \omega = 0,\ \lim_{n\to\infty} ((n-1)\mathbf{I}_{[0,\frac{1}{n}]}(0))$$
$$= \lim_{n\to\infty} n - 1 = \infty \neq 0$$
Therefore $B = \{0\}$.
$P(\{0\}) = 0 \implies X_n \to X\ a.s.(P)$ as $n \to \infty$

□

(c) Since $P = m$ is a probability measure $P(\Omega) = 1 < \infty$. Thus, $X_n \to Xa.s.(P)$ as $n \to \infty \implies X_n \to X$ in probability as $n \to \infty$.

(d) **For** $p = 1$

Proof. We know from (b) that $X_n \to X a.s(P)$.
X is a constant function so it is clearly measurable.
For each n, X_n is a linear combination of a constant function (1), and an indicator function $\mathbf{I}_{[0,\frac{1}{n}]}$, it is a measurable function
Thus, $\{X_n\}_{n\geq 1}$ is a sequence of measurable functions.
$\int X dm = \int 1\mathbf{I}_{[0,1]} dm = 1m([0,1]) = 1 < \infty$
$\int X_n dm = \int 1\mathbf{I}_{[0,1]} + (n-1)\mathbf{I}_{[0,\frac{1}{n}]} dm$
$= 1m([0,1]) + (n-1)m([0,\frac{1}{n}]) = 1 + (n-1)\frac{1}{n} = 2 - \frac{1}{n} < \infty$
Thus, by Scheffe's theorem, $X_n \to^{L_1} X$ □

For $0 < p < 1$:

Proof. Since, $X_n \to^{L_1} X$, $X_n \to^{L_p} X$ □

For $1 < p < \infty$:

Proof. For any $\epsilon > 0$, as $n \to \infty$, $\int |X_n|^{1+\epsilon} dm = \int X_n^{1+\epsilon} dm = \infty$
Thus, $X_n \not\to^{L_p} X$ when $1 < p < \infty$. □

(e) From (d), we know that $X_n \in L^1([0,1], \mathcal{B}([0,1]), m)$ that $X_n \to^{L^1(\mu)} X$.
Thus, $\{X_n\}_{n\geq 1}$ is UI by Problem 2.36.